Google Cloud

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

a. Participant name:
b. Problem Statement:

Participant name: Team EduPulse
Problem Statement: Student Performance Analytics using AlloyDB + Gemini NL Queries

Brief about the idea

EduPulse is an AI-powered student performance analytics platform that transforms how educators interact with academic data. Teachers currently spend 5+ hours per week on manual data analysis, and 1.2 million students drop out annually in India alone due to undetected academic struggles. EduPulse stores comprehensive student performance data in AlloyDB — grades, attendance, assignment scores, exam results across 200 students, 15 subjects, and 4 semesters. Educators simply ask questions in plain English: "Which students are at risk of failing math?" or "Show me the top improvers this semester." Gemini converts these natural language queries into optimized SQL, executes them against AlloyDB, and returns both raw data and an AI-generated natural language summary. This democratizes data access — no SQL knowledge required — enabling teachers to identify at-risk students in seconds instead of hours.

Meeting the Build Criteria

- How did you navigate the requirements of your chosen track to build a custom solution - using ADK, MCP, or AlloyDB AI that effectively addresses the 'What You Must Build' criteria?

We built a complete AlloyDB-centric solution with a custom student performance dataset (200 students, 15 subjects, 2,388 enrollments, 159 risk alerts) — entirely different from any lab default dataset. The schema includes 4 tables with 12 optimized indexes including composite indexes for common query patterns. AlloyDB AI natural language is enabled through Gemini integration: the app sends the complete schema context to Gemini, which generates SELECT-only SQL queries. We enforce strict SQL safety — rejecting any DELETE, DROP, UPDATE, INSERT, or TRUNCATE statements via regex validation. The NL-to-SQL pipeline includes a second Gemini call that summarizes raw query results into teacher-friendly insights. Connection pooling via ThreadedConnectionPool ensures O(1) connection acquisition under concurrent load.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

Differentiation: Existing student information systems require SQL expertise or pre-built reports. EduPulse is the first to combine AlloyDB's performance with Gemini's NL understanding for ad-hoc student queries. Teachers ask any question — no dashboards to configure, no reports to build.

Problem Solution: A teacher wonders which students are struggling. Instead of exporting CSVs and making pivot tables, they type "Students with attendance below 60% and grades below 5.0" and instantly see the at-risk list with an AI summary recommending intervention strategies.

USP: Zero-SQL student analytics. Custom dataset with 2,388 enrollment records across 4 semesters. Dual-AI pipeline (NL→SQL + Results→Summary). SQL injection prevention with SELECT-only enforcement. Sub-3-second end-to-end query response. Proactive at-risk identification saving 5+ teacher hours per week.

List of features offered by the solution

1. Natural Language Queries — Ask any question about students in plain English, no SQL required
2. Gemini NL-to-SQL — Converts questions to optimized SQL using full schema context
3. AI-Powered Summaries — Raw query results summarized into teacher-friendly natural language insights
4. Custom Dataset — 200 students, 15 subjects, 2,388 enrollments, 159 risk alerts across 4 semesters
5. SQL Safety Enforcement — SELECT-only validation rejects all destructive SQL operations
6. Connection Pooling — ThreadedConnectionPool for O(1) connection acquisition under load
7. At-Risk Detection — Instantly identify students with low grades, poor attendance, or declining trends
8. Suggested Queries — 5 pre-built query pills for common analytics questions
9. SQL Transparency — Shows generated SQL alongside results for learning and verification
10. Neo-Academic Dashboard UI — Constellation-themed interface with data table, syntax highlighting
11. AlloyDB Optimized — 12 indexes including composite indexes for sub-second query execution

Process flow diagram or Use-case diagram

PROCESS FLOW:

[Teacher/Admin] → Types natural language question about student performance
↓
[Flask Server] → POST /query endpoint receives {"question": "..."}
↓
[Schema Context Builder] → Prepends full AlloyDB schema to Gemini prompt
↓
[Gemini NL-to-SQL] → Converts natural language to SELECT SQL query
↓
[SQL Validator] → Regex check: rejects DELETE/DROP/UPDATE/INSERT/TRUNCATE
↓
[AlloyDB Execution] → Executes validated SQL via connection pool → returns rows
↓
[Gemini Summarizer] → Second AI call: summarizes raw results into insights
↓
[Response] → Returns: {sql, results[], summary, row_count, execution_time}
↓
[Frontend Display] → Syntax-highlighted SQL, data table, AI insight card

Architecture: Client → Cloud Run → Flask → Gemini (NL→SQL) → AlloyDB → Gemini (Summary) → Response

Database: AlloyDB for PostgreSQL with 4 tables, 12 indexes, 2,388+ records

Technologies to be used in the solution

- AlloyDB for PostgreSQL — AI-ready managed database with optimized indexing
- Gemini 2.0 Flash — Dual-purpose: NL-to-SQL conversion + result summarization
- Google Cloud Run — Serverless deployment with auto-scaling
- psycopg2 — PostgreSQL adapter with ThreadedConnectionPool for connection reuse
- Flask + Flask-CORS — Lightweight HTTP server
- Gunicorn — Production WSGI server with 2 workers, 4 threads each
- Docker — Containerized deployment
- Cloud Build — CI/CD pipeline
- google-genai — Google GenAI SDK for Gemini API access
- HTML5/CSS3/Vanilla JS — Neo-academic constellation-themed frontend
- Google Fonts (Syne, IBM Plex Sans) — Typography
- pytest — Testing framework with 18 passing test cases

Snapshots of the prototype

The prototype features a neo-academic data observatory interface with:

- Deep navy (#0b1120) background with animated constellation dot-grid pattern
- Pulsing "knowledge query portal" search bar with glowing border effect
- 5 suggested query pill buttons for common analytics questions
- Orbital ring CSS loading animation during query processing
- Syntax-highlighted SQL display showing the generated query
- Beautiful data table with sticky headers, alternating row colors, hover effects
- AI Insight card with purple accent showing natural language summary
- Query execution time and row count statistics
- Stats footer: "200 Students | 15 Subjects | 2,388 Records | 4 Semesters"
- Fully responsive design for classroom projector and mobile use

Cloud Run URL: [To be deployed]

Google Cloud H2S

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Thank you